Problem 2.1 (MATLAB)
Autocorrelation function

Investigate the MATLAB ***xcorr*** function for estimation of autocorrelation function.
For easy plotting, use the syntax ***[Rxx, lags] = xcorr (x, 'option')***

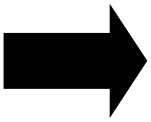
- Define and plot various signals containing components such as, e.g., white noise, filtered white noise, deterministic periodic components, periodic components with random parameters (amplitude, frequency, phase) etc.
- Calculate, plot and interpret the corresponding autocorrelation functions
- Verify by figure inspection the generic properties of autocorrelation functions
- For noise components, try various noise distributions
- Investigate the influence of the 'option' parameter setting for the estimates for autocorrelation functions

Problem 2.2 (MATLAB)
Crosscorrelation function

Investigate the MATLAB ***xcorr*** function for estimation of crosscorrelation function between two signals. For easy plotting, use the syntax ***[Rxy, lags] = xcorr (x,y,'option')***

- Define in different ways two correlated signals, for example a sent signal and a received signal that is an attenuated, delayed and noisy version of the sent signal
- Calculate, plot and interpret the corresponding crosscorrelation functions
- Verify by figure inspection the generic properties of crosscorrelation functions
- Swap the roles of the signal x and y
- Investigate the influence of the 'option' parameter setting for the estimates for crosscorrelation functions

Theoretical problems



Problem 2.3 (Theoretical + optionally MATLAB)
Calculation of ensemble and estimated autocorrelation functions

A stochastic signal is given by

$$X(t) = A\cos(\omega t + \theta), \quad A = A_0,\ \omega = \omega_0,\ \theta = Uniform[0; 2\pi]$$

1) Sketch the pdf (probability density function) for amplitude, frequency and phase

2) Sketch the ensemble (the family) $X(t)$

3) Find the average value of the ensemble $\mu_X = E[X(t)]$

4) Find the average time value $\hat{\mu}_X = A[X(t)]$

5) Find the autocorrelation function $R_{XX}(\tau) = E[X(t)X(t+\tau)]$ (ensemble averaging)

6) Find the autocorrelation function $\hat{R}_{XX}(\tau) = A[X(t)X(t+\tau)]$ (time averaging)

7) Is the signal $X(t)$ WSS, weak-sense stationary?

8) Is it ergodic?

9) Check the characteristics of the autocorrelation function $R_{XX}(\tau)$

10) Find and interpret $E[X^2(t)], Var[X(t)]$ and the rms-value of the signal

Problem 2.4 (Theoretical and optionally MATLAB)
Calculation of ensemble and estimated autocorrelation functions

A stochastic signal is given by

$$X(t) = A\cos(\omega t + \theta), \quad A = N(\mu_a, \sigma_a{}^2),\ \omega = \omega_0,\ \theta = Uniform[0; 2\pi] \qquad A, \theta\ independent$$

1) Sketch the pdf (probability density function) for amplitude, frequency and phase

2) Sketch the ensamble (the family) $X(t)$

3) Find the average value of the ensamble $\mu_X = E[X(t)]$

4) Find the average time value $\hat{\mu}_X = A[X(t)]$

5) Find the autocorrelation function $R_{XX}(\tau) = E[X(t)X(t+\tau)]$ (ensemble averaging)

6) Find autocorrelation function $\hat{R}_{XX}(\tau) = A[X(t)X(t+\tau)]$ (time averaging)

7) Is the signal $X(t)$ WSS, weak-sense stationary?

8) Is it ergodic?

9) Check the characteristics of the autocorrelation function $R_{XX}(\tau)$

10) Find and interpret $E[X^2(t)], Var[X(t)]$ and the rms-value of the signal